

Design and Simulation of Sinusoidal Assist Pattern Generator for Robotic Gait-Training Walker Based on Physical Therapist Force Profiles

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Abstract—This study presents the design and simulation of an assist pattern generator for a pelvic-control robotic gait-training walker, using a sinusoidal model informed by experimentally measured physical therapist (PT) assistance forces. In our previous experiment, three-dimensional pelvic assistance forces applied by PTs during hemiplegic gait were quantified, identifying characteristic peak timings and amplitudes for forward and upward components of the left and right hands. These findings were used to define the parameters of half-wavelength sinusoidal functions, which served as the basis for the assist pattern generator. The model was implemented in MATLAB/Simulink and evaluated under different gait phase timing conditions. Results show that the generated force profiles closely matched the clinical measurements in both amplitude and timing, supporting the feasibility of translating PT strategies into robotic control algorithms. This approach provides a quantitative pathway for embedding therapist-like assistance patterns into rehabilitation devices, potentially improving training outcomes for patients with gait impairments.

Index Terms—Gait rehabilitation, robotic walker, assistive force, pelvic control, sinusoidal model

I. INTRODUCTION

Japan is rapidly transitioning to a super-aged society, with individuals aged 65 and over comprising 29.1% of the population in 2023—a figure projected to reach 38.7% by 2070 [1]. This demographic transformation has resulted in escalating medical and long-term care costs, making the extension of healthy life expectancy a critical national priority. Gait ability, which is fundamental to Activities of Daily Living (ADL), is intrinsically linked to maintaining physical, mental, and social well-being.

Stroke represents a leading cause of long-term disability in adults, often leaving patients with significant gait impairments [2]. In post-stroke rehabilitation, sufficient volumes of high-quality gait training are known to promote neural reorganization and functional recovery [3]. However, delivering such training is frequently challenging due to shortages of physical therapists (PTs) and clinical time constraints imposed by healthcare systems [4].

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To address these limitations, Robot-Assisted Gait Training (RAGT) has emerged as a promising intervention. RAGT enables consistent, high-intensity, and repetitive walking practice, with devices such as the Lokomat®, G-EO System, and HAL demonstrating clinical effectiveness [5]. However, these systems are typically large-scale and expensive, confining their use primarily to hospital settings and thus limiting patient access and training duration. Furthermore, exoskeleton-type robots that control the entire lower extremities have been criticized for potentially constraining natural pelvic motion and enforcing uniform gait patterns that may not align with individual patient needs [6].

In response to these challenges, we propose a novel approach: a pelvic-assist gait support robot integrated with a walker platform. Walkers are widely utilized in clinical practice because their four-point support structure enhances stability by widening the base of support, while also reducing the load on the lower limbs by distributing weight [7]. Compared to large-scale robotic systems, this approach offers the potential for reduced costs and improved accessibility.

The innovation of our research lies in specifically targeting the pelvis for assistance. The pelvis serves as the anatomical link between the trunk and lower extremities, playing a crucial role in gait efficiency and stability through its complex three-dimensional movements: rotation, lateral obliquity, and anterior-posterior tilt [8]. Unlike conventional robots that rigidly control the entire lower extremities, providing assistance at the pelvis as a proximal control point respects individual gait patterns and promotes the acquisition of more natural movement strategies [9].

Building upon previous work, we have focused on analyzing the skilled manual techniques employed by expert physical therapists, specifically measuring and characterizing the three-dimensional forces and moments that experienced PTs apply to patients' pelvises during gait assistance.

The objective of this study is to translate the tacit knowledge embedded in PT assistance techniques into explicit assistance patterns suitable for robotic implementation. By modeling these patterns using sinusoidal functions, we aim to achieve

low computational overhead and enable real-time control. This approach is expected to contribute to the development of a practical, walker-based robotic system that patients can utilize in their daily environments without temporal or spatial constraints.

II. ANALYSIS OF PELVIC ASSISTANCE BY PHYSICAL THERAPISTS (OUR PREVIOUS STUDY)

A. Maintaining the Integrity of the Specifications

We published a paper entitled "Evaluating pelvic external forces and moments in assisted walking: implications for the development of robotic-assisted walkers" to quantify the external forces and moments applied to the pelvis by expert physical therapists (PTs) and to elucidate their periodic characteristics [10]. The following presents a summary of this study.

This study aimed to quantify the external forces and moments that expert physical therapists (PTs) apply to the pelvis during gait assistance and to identify their cyclical patterns. Two healthy adult males participated as subjects. They simulated hemiplegic gait by limiting the weight-bearing load on their right leg to 70% of their body weight while walking on a treadmill with a walker. Eight experienced PTs provided manual assistance, and the three-dimensional forces and moments applied to the pelvis were measured using a sensor-equipped pelvic belt. These data were analyzed to determine the periodicity and consistency of assistance patterns throughout the gait cycle.

The analysis revealed distinct and reproducible patterns in the assistance provided by the PTs. Four characteristic assistance techniques were observed with high reproducibility. First, during the right foot contact phase (approximately 19% of the gait cycle), the right hand applied an anterior force to the pelvis, facilitating smooth weight transfer onto the right leg. Second, at the initiation of the left leg's swing phase (30%), the left hand provided upward assistance to the pelvis to ensure adequate toe clearance. Third, during the late swing phase of the left leg (44%), the left hand applied a rotational moment to the pelvis, promoting forward advancement of the left pelvis and thereby increasing step length. Fourth, even during the swing phase of the unrestricted right leg (79%), subtle upward assistance was provided by the right hand to support overall gait stability.

These assistance techniques resulted in increased loading on the subjects' right legs and improved temporal symmetry between the stance and swing phases of both legs. Furthermore, gait symmetry was enhanced, contributing to a more efficient and stable walking pattern.

This research successfully quantified the sophisticated manual techniques employed by expert PTs in pelvic assistance, detailing the specific magnitudes and temporal characteristics of the applied forces. Based on these findings, we developed and validated a gait assistance pattern generator through computational simulation.

III. DESIGN OF THE ASSISTANCE PATTERN GENERATOR

A. Definition and Parameter Setting of Assistance Patterns

Assistance patterns were selected based on the forces that demonstrated high periodicity ($APCC \geq 0.6$) in the results of our previous study. The forward force applied by the left hand was defined as a substitute for the pelvic rotational moment applied by the left hand, as discussed in our previous study. In this definition, the peak timing was set to correspond to that of the pelvic rotational moment, and the peak amplitude was hypothetically determined considering the balance with other assistance forces. A unimanual rotational moment contributes little to pelvic motion on its own; therefore, in the control strategy, assistance was limited exclusively to three-dimensional forces.

The selected assistance forces were: forward force applied by the right hand, upward force applied by the left hand, forward force applied by the left hand, and upward force applied by the right hand. The parameters of these assistance patterns are defined in Table I.

Although the actual assistance waveforms applied by PTs are complex, their essential characteristic lies in the periodic application of forces: they begin at a specific timing, reach a peak, and then diminish [10]. To capture this feature while maintaining low computational load suitable for real-time control, a half-wave sinusoidal function was adopted as the fundamental assistance waveform.

B. Mathematical Modeling Using Sinusoidal Functions

The assistance force $F(t)$ is defined as a function of normalized gait cycle time t ($0 \leq t \leq 1$). During the assistance interval, $T_{\text{start}} \leq t \leq T_{\text{end}}$, a half-wavelength sinusoidal function is generated, as expressed in (1).

$$F(t) = \begin{cases} A \sin\left(\frac{\pi}{\Delta T} (t - T_{\text{start}})\right), & T_{\text{start}} \leq t \leq T_{\text{end}}, \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

Here, A denotes the peak assistance force, and $\Delta T = T_{\text{end}} - T_{\text{start}}$ represents the **assistance duration** (in this study, $\Delta T = 0.20$, corresponding to 20% of the gait cycle).

The start and end timings of the assistance are defined by (2):

$$T_{\text{start}} = T_{\text{peak}} - \frac{\Delta T}{2}, \quad T_{\text{end}} = T_{\text{peak}} + \frac{\Delta T}{2} \quad (2)$$

TABLE I
FOUR SELECTED ASSISTANCE FORCES

Assistance pattern	Peak amplitude (A) [N/kg]	Peak timing (T_{peak}) [%GC]
a. Right hand forward	0.175	19
b. Left hand upward	0.475	30
c. Left hand forward	0.150	44
d. Right hand upward	0.212	79

%GC = % of Gait Cycle.

The peak timing T_{peak} was determined for each assistance pattern according to the targeted phase within the gait cycle.

C. Simulation of Generated Patterns

The behavior of the designed assistance pattern generator was verified through simulation. Fig. 1 shows the generation results of four types of assistive forces over a right gait cycle (0–100%).

In Fig. 1, the horizontal axis represents the gait cycle (%), and the vertical axis represents the body-weight normalized assistive force (N/kg). The four assistive forces (right-hand forward force, left-hand upward force, left-hand forward force, and right-hand upward force) are plotted as half-wave sinusoids, and each peak is annotated with its timing and value (e.g., a. 19% (0.175 N/kg)). At the bottom of the graph, stance and swing phases of the right and left legs, derived from the assisted condition in our previous study, are shown. The legend indicates which line type corresponds to each assistive pattern.

This simulation confirmed that the designed algorithm was able to generate assistive patterns according to the parameters presented in Fig. 1. Each assistive force was designed to act at critical phases of the gait cycle.

Specifically, the **a.** right-hand forward force acts during the loading response phase of the right leg, promoting forward weight transfer. The **b.** left-hand upward force and the **c.** left-hand forward force act during early and late swing of the left leg, respectively, and are considered to facilitate leg swing. These timings are consistent with the phases in which physical therapists' assistance intentions have been suggested in previous studies [10].

IV. SIMULATION VERIFICATION OF THE ASSISTANCE PATTERN GENERATOR

We evaluated the effects of a simplified assistance force model, derived from quantitative analysis of support behaviors by physical therapists (PTs), on the kinematics of a human model through simulation. Using a six-link rigid body human model with a height of 170 cm and weight of 65 kg developed

in previous study, we compared three-dimensional movements of the hip joint, knee joint, and pelvis with and without external forces [11].

A. Human Model and Simulation Environment

The simulations were conducted on MATLAB/Simulink (MathWorks, Inc.). All connections between the pelvis, thigh, and shank were modeled as one-degree-of-freedom 'Revolute Joints', and the hip joint axis was offset 0.10 m laterally from the pelvic center of mass. The initial posture was set at full hip extension. At the start of the simulation, a hip flexion torque was applied only for the first 0.08 s, after which the motion was continued solely by the spring–damper properties and inertia of the joints. The gravitational acceleration was set to 9.81 m/s^2 , and one gait cycle was normalized to 1 s. The period from 0.15 to 0.42 s (15–48.8%GC) was defined as the swing phase of the left leg.

B. Implementation of Assistance Force Patterns

The assistive forces applied by the left hand during the swing phase were limited to two types: upward (LH_Up) and forward (LH_Fwd), as shown in Fig. 2. Each force was modeled as a half-wave sinusoid acting for 20% of the gait cycle ($\Delta T = 0.20 \text{ s}$). The angular frequency of the half-wave sinusoid was constant, given by

$$\omega = \frac{\pi}{\Delta T} = \frac{\pi}{0.20} = 15.708 \text{ rad/s}.$$

Peak Timing and Duration

The swing phase was redefined with its onset at 0 s. The peak timings were set at 0.17 s for LH_Up and 0.42 s for LH_Fwd. Accordingly, the onset (t_{start}) and termination (t_{end}) times of each half-wave were given by

$$t_{\text{start}} = t_{\text{peak}} - \frac{\Delta T}{2}, \quad t_{\text{end}} = t_{\text{peak}} + \frac{\Delta T}{2},$$

$$\text{LH_Up: } t_{\text{peak}} = 0.17 \text{ s}, \quad t_{\text{start}} = 0.07 \text{ s}, \quad t_{\text{end}} = 0.27 \text{ s},$$

$$\text{LH_Fwd: } t_{\text{peak}} = 0.42 \text{ s}, \quad t_{\text{start}} = 0.32 \text{ s}, \quad t_{\text{end}} = 0.52 \text{ s}.$$

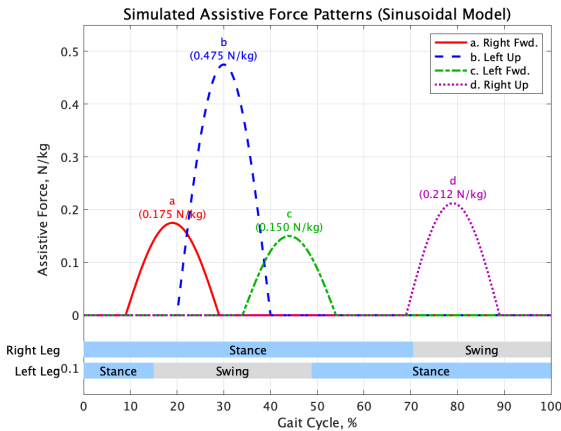


Fig. 1. Simulated assistive force patterns based on the sinusoidal model.

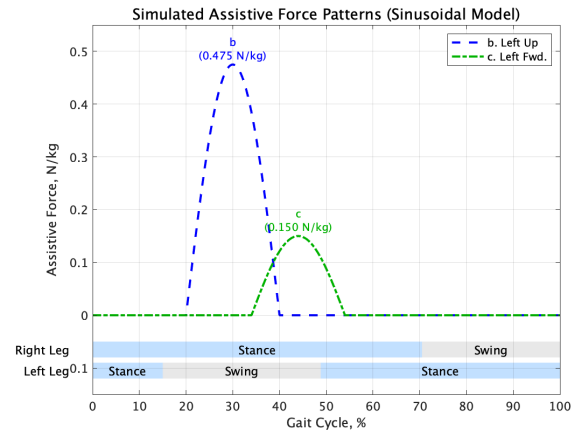


Fig. 2. Upward and forward forces applied during the swing phase.

TABLE II
ASSISTIVE FORCE PARAMETERS (65 KG MODEL)

Force	A_{abs} [N]	t_{peak} [s]	$t_{start}-t_{end}$ [s]	ϕ [rad]
LH_Up	30.875	0.17	0.07–0.27	–1.10
LH_Fwd	9.75	0.42	0.32–0.52	–5.03 ($\equiv$ 1.26)

Waveform

$$F(t) = \begin{cases} A_{abs} \sin[\omega(t - t_{start})], & t_{start} \leq t \leq t_{end}, \\ 0, & \text{otherwise,} \end{cases}$$

$$\phi = \frac{\pi}{2} - \omega t_{peak}$$

where ϕ was set as the phase offset in the Sine Wave block.

Simulink Implementation

- **Pulse Generator:** Period = 1 s, Pulse width = 20%, Phase delay = t_{start} (0.07 s / 0.32 s)
- **Sine Wave:** Amplitude = A_{abs} , Frequency = ω , Phase = ϕ (see Table II)
- The gating signal and the sinusoidal wave were multiplied using a Product block, and the Outport block was set to “Disable → reset to 0” to ensure that the output was completely zero outside the gating interval.

Through this implementation, the LH_Up force was applied as a single half-wave during 0.07–0.27 s of the swing phase, and the LH_Fwd force was applied as a single half-wave during 0.32–0.52 s. Thus, the designed assistive pattern was accurately reproduced.

C. Data Acquisition and Processing

Kinematic data, including the joint angles of the hip and knee and the three-dimensional orientation of the pelvis, were obtained from the simulation. The data were exported to the MATLAB workspace using the ‘To Workspace’ block. Subsequently, they were processed using the ‘simulationOutput.m’ script. Specifically, the joint angles were converted from radians to degrees, and the segment corresponding to the swing phase (data points 20 to 420) was resampled into an array of 1000 normalized samples for comparative analysis.

V. RESULTS

Simulations were conducted under two conditions: applying four types of simplified assistive forces (“With Force”, WF) and applying no external forces (“Without Force”, WOF).

Fig. 3 compares, during the normalized swing phase, the upward assistive force (blue, dash-dot), the forward assistive force (red, dash-dot), and the flexion angles of the hip (green; positive values indicate flexion) and knee (cyan; negative values indicate flexion). The solid lines (WF) and the dashed lines (WOF) were nearly identical for both joints, indicating that the applied assistive forces had little effect on the overall angular displacements of the hip and knee.

Fig. 4 shows the comparison of the three rotational degrees of freedom of the pelvis. Pelvic depression on the swing side was reduced under WF (blue, solid) compared with WOF (purple, dashed). A slight reduction was also observed in right

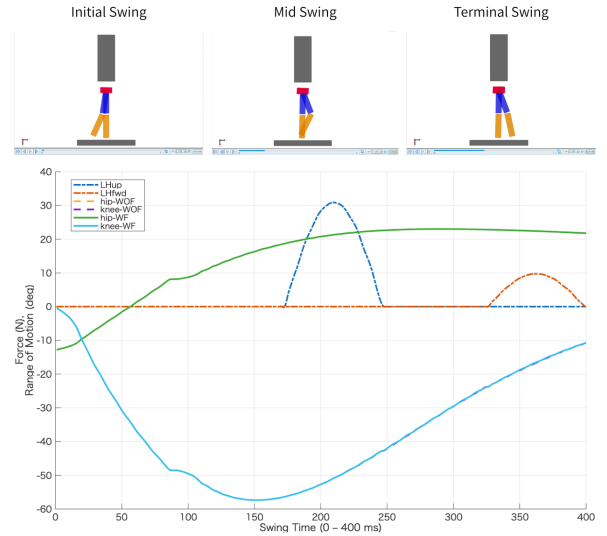


Fig. 3. Assistive force generated by the sinusoidal model and flexion angles of the hip and knee.

pelvic rotation under WF (orange, solid) relative to WOF (green, dashed). In contrast, pelvic anterior tilt exhibited no substantial change in its kinematic pattern.

VI. CONCLUSION

In this simulation, the simplified sinusoidal assistive forces induced only minor changes in the motion of the human body model. When compared with the results of our previous studies, these findings provide important insights for the development of robotic gait-assist devices [11].

This study designed an assistive pattern generator and evaluated the effects of simplified assistive forces on human body model kinematics through simulation. The proposed sinusoidal mathematical model successfully reproduced both the peak timings and amplitudes of assistance observed in our previous study, effectively capturing the essential characteristics of therapist-provided assistance. This demonstrates the feasibility

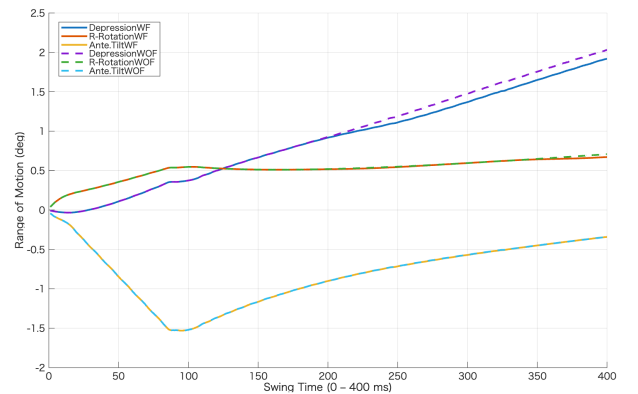


Fig. 4. Comparison of pelvic motion in three planes.

of generating external force input patterns for the pelvis based on quantitative data.

In this study, gravity was included in the simulation environment; however, voluntary muscular effort was not modeled. As a result, the applied assistive forces may have been counteracted by gravitational effects, reducing their apparent influence. To better isolate the effect of external assistance, future work will examine simulations under conditions where gravity is removed, thereby allowing us to evaluate assistive forces independently of gravitational loading.

The principal contribution of this study is demonstrating that a computationally efficient, simplified mathematical model can generate appropriate assistive forces in accordance with gait phase. This represents a significant and practical step toward translating the tacit knowledge of physical therapists into engineering models, bringing the development of adaptive robotic gait-assist devices closer to clinical application.

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